

Diwaakar Jayaprakash

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SUMMARY

Mechanical & Systems Engineer specializing in CAD modeling (SolidWorks, Siemens NX, Fusion 360), ANSYS thermal/structural FEA, and precision sheet metal/enclosure design aligned with DFM/DFA and IP/IEC standards. Combines virtual simulation, tolerance stack-ups, and embedded edge systems (C/C++, Python, Linux) to deliver production-ready electromechanical hardware.

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram, India)

Bachelor of Technology in Mechanical Engineering

Expected May 2027 (2023 – Present)

Relevant Coursework: Prototyping and Testing, Kinematics and Dynamics of Machinery & Multi-body system Dynamics, Design of Machine elements, Sustainable Manufacturing, Manufacturing Processes, Heat Transfer, Materials Science, Thermodynamics, Fluid Mechanics.

EXPERIENCE

Mechanical Engineering Intern (Manufacturing & DFM)

Aug 2026 – Present

Ohm Energy Management Systems Pvt Ltd (Al Musandam Group)

Chennai, India

- Audited electrical enclosure fabrication workflows across CNC punching, laser cutting, hydraulic bending, and 7-tank chemical pre-treatment lines.
- Standardized manufacturing SOPs across sheet metal assembly and powder coating, eliminating routing bottlenecks and tightening dimensional repeatability.

Hardware & IoT Engineering Intern

Jun 2026 – Aug 2026

REConnect Energy Solutions Limited

Bengaluru, India

- Conducted structural and thermal DFM for an IP54-rated Gridbox enclosure, validating compliance against IEC 60529 (Ingress) and IEC 61000 (EMC/ESD) standards.
- Configured automated edge deployment on Debian 12 nodes via Bash and SoftEther VPN, implementing systemd watchdog routines for automated node recovery.
- Monitored field telemetry across 600+ substation sites; optimized data ingestion using SQL and wrote ESP32 firmware in C++ for low-latency MQTT communications.

Mechanical Engineering Intern

Jun 2025 – Aug 2025

Tech Mahindra (Client: ASML)

Bengaluru, India

- Monitored high-throughput semiconductor lithography processes (>1,000 units/min) and engineered a MATLAB/Python automated data pipeline for critical dimension defect detection.
- Performed 3D packaging constraint audits and 1D/3D GD&T tolerance stack-ups in Siemens NX to eliminate subsystem mechanical interference.

Research Assistant – EV Safety & Battery Systems

Mar 2025 – Apr 2025

IIITDM Kancheepuram

Chennai, India

- Conducted transient thermal FEA on Li-ion battery packs using ANSYS Thermal, correlating models against infrared thermography data across variable discharge rates.
- Fabricated and instrumented benchtop test fixtures to evaluate passive cooling and thermal runaway containment geometries.

ENGINEERING PROJECTS

Mars Research Station (MaRS) Rover – Team Shunya

Apr 2024 – Present

Mechanical Lead & Subsystems

- Designed an autonomous sampling module with an indexed 6-beaker carousel, lead-screw core drill, and assay dispensing unit using SolidWorks; integrated custom dust-tight labyrinth seals and high-strength FDM components.
- Engineered a modular rocker-bogie passive suspension chassis; validated dynamic terrain stability and ground-contact normal forces via SolidWorks Motion Simulation.
- Directed mechanical integration and structural load audits for a 6-DoF manipulator arm; team placed 4th globally at ERC Remote 2025 (Poland), 10th at IRC 2026, and 16th at IRC 2025.

Closed-Loop EEG Cognitive Fatigue Headset

Feb 2026 – Apr 2026

Lead Hardware & Prototyping Designer

- Iterated 3 generations (V1–V3) of ergonomic headband assemblies, optimizing spring-loaded dry electrode contact pressure while minimizing overall mass.
- Isolated rigid PCBAs from external headband flex using elastomeric dampeners to prevent mechanical strain and biometric motion artifacts; validated frame stress distributions through FEA and 2-hour wear tests.

TECHNICAL SKILLS

- CAD & 3D Modeling:** Autodesk Fusion 360, SolidWorks (Part, Assembly, Motion Analysis), Siemens NX, AutoCAD, Rhinoceros 3D, Onshape
- FEA & Thermal Simulation:** ANSYS Thermal, CFD Fundamentals, Kinematic Analysis, Tolerance Stack-up Analysis
- Programming, Data & Web:** Python (NumPy, Pandas), MATLAB & Simulink, C/C++, MySQL, HTML5, CSS3, JavaScript
- Hardware & Prototyping:** Rapid Prototyping (3D Printing / DfAM), CNC Machining, Laser Cutting, Edge IoT Integration
- Domains & Standards:** Robotics & Electromechanical Systems, Semiconductor Process Engineering, IoT Hardware Packaging, Wearable Electronics, Digital Twins, IEC 60529 (IP54/IP67), IEC 61000

AWARDS & COMPETITIONS

- European Rover Challenge (ERC) Remote 2025:** 4th Globally (Field of 50+ international teams)
- International Rover Challenge (IRC) 2026:** 10th Globally
- International Rover Challenge (IRC) 2025:** 16th Globally (Field of 226 international teams)